

Quantum order by disorder at the multiphase point of a bilayer transverse-field ANNNI model

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We study the transverse-field axial next-nearest-neighbor Ising (ANNNI) model on the square lattice and on a bilayer of two such layers, with stochastic series expansion quantum Monte Carlo at temperature $T = 0.15 J_0$. At the classical multiphase point $\kappa = J_2/J_0 = 1/2$, every configuration of straight domain walls that separates domains of at least two sites has the same energy. Second-order perturbation theory in the transverse field $g = \Gamma/J_0$ shows that quantum fluctuations lift this degeneracy in favor of the $\langle 3 \rangle$ structure $\uparrow\uparrow\downarrow\downarrow$, which occupies a wedge of width $9g^2/[4(2+r)(3+r)(5+r)]$ around $\kappa = 1/2$, where r is the interlayer coupling in units of J_0 ; the interlayer coupling narrows the wedge by a factor 2.4 between $r = 0$ and $r = 1$. Local and cluster updates freeze in the modulated region, and annealing from the paramagnet locks into $\langle 3 \rangle$ even where the antiphase $\langle 2 \rangle$ is stable. We therefore follow the competing ordered states as separate branches and compare their free energies by thermodynamic integration in the field. **[Results: location of the $\langle 3 \rangle$ wedge from the branch free energies and comparison with the $O(g^2)$ prediction; the paramagnetic boundary at $r = 0$ and $r = 1$; bounds on the modulated region.]**

I. INTRODUCTION

The axial next-nearest-neighbor Ising (ANNNI) model is the simplest lattice model in which competing ferromagnetic and antiferromagnetic interactions produce spatially modulated order [1, 2]. Along one axis, a ferromagnetic nearest-neighbor coupling J_1 competes with an antiferromagnetic next-nearest-neighbor coupling J_2 . At zero temperature the ferromagnet is stable for $\kappa = J_2/J_1 < 1/2$ and the antiphase $\langle 2 \rangle$, $\uparrow\uparrow\downarrow\downarrow$, for $\kappa > 1/2$, and at the multiphase point $\kappa = 1/2$ a large number of structures are degenerate. In the three-dimensional classical model, thermal fluctuations lift this degeneracy and generate an infinite sequence of commensurate phases that spring from the multiphase point [2–4]. In two dimensions the modulated region is instead dominated by a floating incommensurate phase with algebraic correlations [5, 6], whose extent was settled only recently [7–9].

A transverse field Γ adds quantum fluctuations. The transverse-field ANNNI chain is related to the classical two-dimensional model by the Suzuki–Trotter mapping [10] and has been studied in great detail [11–14]; the width of its floating phase was long disputed [12, 13]. In two and more dimensions the quantum ANNNI model has been treated only by mean-field and random-phase approximations [15], by large-spin perturbation theory for an anisotropic Heisenberg analog [16, 17], and by field theory of the quantum Lifshitz point [18]. To our knowledge there is no unbiased numerical study of the transverse-field ANNNI model with axial frustration on the square lattice or on its bilayer extension. Re-

lated frustrated transverse-field Ising models with diagonal couplings have been studied with quantum Monte Carlo [19–21] and, for bilayers, with cluster mean-field theory [22–24]; frustrated transverse-field Ising models are also realized on programmable quantum annealers [25].

Here we consider two square layers with axial frustration, coupled by an interlayer exchange, and ask two questions. What selects the ordered state near the multiphase point once quantum fluctuations are present, and how does the second layer change the answer? At the multiphase point of the square-lattice model, the in-plane ferromagnetic coupling perpendicular to the frustrated axis forces domain walls to be straight lines, so the degeneracy is subextensive and walls cannot move at low order in Γ . This makes the low-field problem accessible to perturbation theory. At second order in Γ we find that quantum fluctuations select the $\langle 3 \rangle$ structure $\uparrow\uparrow\downarrow\downarrow$ in a wedge around $\kappa = 1/2$ whose width grows as Γ^2 , and that the interlayer coupling narrows the wedge. This differs both from the one-dimensional chain, where walls are mobile and the degeneracy gives way to a floating phase [11], and from the large-spin Heisenberg analog, where a sequence $\langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \dots$ appears on the ferromagnetic side [16, 17].

We test these predictions with stochastic series expansion (SSE) quantum Monte Carlo [26–28] at $T = 0.15 J_0$. Frustration makes the modulated region hard to equilibrate: cluster updates percolate at small field, and annealing from the paramagnet selects the modulation that forms at the ordering transition and cannot relax to a different commensurate structure at lower field. We therefore follow each competing ordered state as a separate branch and compare the branches by their free energies, obtained by thermodynamic integration in the field.

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[Summary of results.]

Section II defines the model and its symmetries. Section III derives the low-field phase diagram near the multiphase point. Section IV describes the SSE simulations, the branch protocol and their validation, and Sec. V presents the results. Section VI concludes.

II. MODEL

A. Hamiltonian

We consider two square layers $\ell = 1, 2$ of $L_x \times L_y$ sites (i, j) with periodic boundaries and $N = 2L_x L_y$ spins,

$$\begin{aligned} \hat{H} = & -J_x \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^z \hat{\sigma}_{i+1,j,\ell}^z - J_y \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^z \hat{\sigma}_{i,j+1,\ell}^z \\ & + J_2 \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^z \hat{\sigma}_{i,j+2,\ell}^z - J_\perp \sum_{i,j} \hat{\sigma}_{i,j,1}^z \hat{\sigma}_{i,j,2}^z \\ & - \Gamma \sum_{i,j,\ell} \hat{\sigma}_{i,j,\ell}^x. \end{aligned} \quad (1)$$

The frustrated axis is y . We set $J_x = J_y = J_0 > 0$ and use

$$\kappa = \frac{J_2}{J_0}, \quad g = \frac{\Gamma}{J_0}, \quad r_\perp = \frac{J_\perp}{J_0}. \quad (2)$$

The single-layer model is the case $r_\perp = 0$, in which the two layers decouple. The off-diagonal matrix elements $-\Gamma$ are nonpositive, so the model is stoquastic and has no sign problem in quantum Monte Carlo.

B. Symmetries

Two unitary transformations relate different couplings exactly. The layer flip $\hat{U}_2 = \prod_{i,j} \hat{\sigma}_{i,j,2}^x$ reverses $\hat{\sigma}^z$ in layer 2, changes the sign of the interlayer term and leaves all other terms invariant, so $\hat{H}(\kappa, g, r_\perp)$ and $\hat{H}(\kappa, g, -r_\perp)$ have the same spectrum and the same in-plane correlations; only the stacking of the two layers is reversed. The alternate-row spin flip $\hat{V} = \prod_{i,j \text{ odd}, \ell} \hat{\sigma}_{i,j,\ell}^x$ (for even L_y) reverses the sign of J_y and leaves J_x , J_2 and $J_\perp$ unchanged. We therefore take $J_y > 0$ and $r_\perp \geq 0$ without loss of generality, and write $r = |r_\perp|$.

C. Classical limit and ordered structures

At $g = 0$ the in-plane coupling J_x aligns neighboring spins along x , so in any low-energy configuration each column along y repeats in x , and the two layers are stacked in parallel for $r_\perp > 0$. The energy per spin is then fixed by the sequence of domains along y . For a structure in which all domains have at least two sites and the wall density along y is ρ (walls per site),

$$e_0 = -2 + \kappa - \frac{r}{2} + (2 - 4\kappa)\rho, \quad (3)$$

in units of J_0 . The ferromagnet (FM, $\rho = 0$) has $e_0 = -2 + \kappa - r/2$, the antiphase $\langle 2 \rangle$ ($\rho = 1/2$) has $e_0 = -1 - \kappa - r/2$, and the structure $\langle 3 \rangle$, $\uparrow\uparrow\uparrow\downarrow\downarrow\downarrow$ ($\rho = 1/3$), has $e_0 = -4/3 - \kappa/3 - r/2$. At $\kappa = 1/2$ the wall energy $2 - 4\kappa$ vanishes and every structure with domains of at least two sites is degenerate. Because walls must be straight lines along x , the number of such structures grows as $\exp(cL_y)$, not $\exp(cN)$: the degeneracy is subextensive.

At zero temperature, the Suzuki–Trotter mapping [10] relates the two-dimensional quantum model to an anisotropic three-dimensional classical ANNNI model, for which the thermal lifting of the multiphase degeneracy is known [3]. At the temperature of our simulations the thermodynamics is two-dimensional. The ferromagnet then orders in the two-dimensional Ising universality class, and the structure $\langle n \rangle$, of period $2n$, carries a $\mathbb{Z}_{2n}$ clock-type order that can melt through a floating phase [5, 6]. In mean-field theory the paramagnet becomes unstable at the wavevector $\cos q^* = 1/(4\kappa)$ for $\kappa > 1/4$ [2]. This is the wavevector of the instability at the ordering transition, not of the low-temperature ordered state.

D. Observables

We measure the energy per spin e , the transverse magnetization $m_x = N^{-1} \sum \langle \hat{\sigma}^x \rangle$, and equal-time zz correlations. Modulated order is characterized by the per-layer axial amplitudes

$$a_\ell(q) = \frac{1}{L_x L_y} \sum_{i,j} \sigma_{i,j,\ell}^z e^{iqj}, \quad q = \frac{2\pi n}{L_y}, \quad (4)$$

and the structure factor $S(q) = \frac{1}{2} \sum_\ell \langle |a_\ell(q)|^2 \rangle$. The dominant wavevector $q^* \in [0, \pi]$ maximizes $S(q)$. Since $a_\ell(-q) = a_\ell(q)^*$, the order parameter of a modulated state with $0 < q^* < \pi$ is $O(q^*) = 2S(q^*)$, and of the ferromagnet $O(0) = S(0)$. Both saturate at 1 for the ferromagnet and for $\langle 2 \rangle$, and at $8/9$ for $\langle 3 \rangle$; we normalize by these saturation values when comparing structures. Transitions are located with Binder cumulants [29], $U_4 = 1 - \langle m^4 \rangle / (3\langle m^2 \rangle^2)$ for the real ferromagnetic order parameter and $U_4 = 2 - \langle |a|^4 \rangle / \langle |a|^2 \rangle^2$ for the complex modulated one, and with the second-moment correlation lengths ξ_x and ξ_y divided by the linear size. We compute ξ_x and ξ_y separately because the couplings are anisotropic. For the same reason we use the crossings of U_4 only to locate transitions: the value of U_4 at the crossing is not universal in anisotropic Ising models with competing interactions [30]. [Confirm the final estimator set once the production analysis is fixed.]

TABLE I. Local field h of Eq. (5) at $\kappa = 1/2$ for spins in straight-wall structures, by position in a domain of n sites.

Position	n	h
Bulk (two or more sites from both walls)	≥ 5	$3 + r$
Either site of a two-site domain	2	$3 + r$
Next to a wall	≥ 3	$2 + r$
Center of a three-site domain	3	$5 + r$
Second site from a wall	≥ 4	$4 + r$

III. LOW-FIELD PERTURBATION THEORY NEAR THE MULTIPHASE POINT

A. Local fields

For $g \ll 1$ and temperatures far below the excitation gaps, the free energy of an ordered structure is its energy (3) dressed by virtual single-spin flips. A flip of spin s costs $2hJ_0$, where h is the local field in units of J_0 . For a straight-wall structure with favorable stacking,

$$h = 2 + r + a_{+1} + a_{-1} - \kappa(a_{+2} + a_{-2}), \quad a_d = s_j s_{j+d}, \quad (5)$$

where j is the coordinate along y and the constant $2 + r$ collects the two in-plane neighbors along x and the partner spin in the other layer. At second order,

$$e_2 = -\frac{g^2}{2N} \sum_{\text{spins}} \frac{1}{h}. \quad (6)$$

At $\kappa = 1/2$, Eq. (5) takes five values, listed in Table I. Every spin in the bulk of a domain, and every spin of a two-site domain, has $h = 3 + r$. A spin next to a wall has the smaller field $2 + r$ and therefore gains more energy from quantum fluctuations. The field of a spin depends only on the length of its own domain, provided every domain has at least two sites.

B. The $\langle 3 \rangle$ wedge

Measured from the ferromagnet, the energy per spin of a sequence of domains of lengths n_k is

$$\Delta e = \frac{\sum_k [(2 - 4\kappa) - \frac{g^2}{2} F(n_k)]}{\sum_k n_k}, \quad (7)$$

where $F(n) = \sum_{\text{domain}} h^{-1} - n/(3 + r)$ is the excess second-order gain of one domain. From Table I,

$$F(2) = 0, \quad F(3) = \frac{6}{(2 + r)(3 + r)(5 + r)},$$

$$F(n \geq 4) = F(3) - \frac{2}{(3 + r)(4 + r)(5 + r)}. \quad (8)$$

Because Eq. (7) is a ratio of sums, its minimum over all sequences is reached by a periodic structure $\langle n \rangle$ with a

single domain length, the one that minimizes the per-domain cost divided by n , or by the ferromagnet if every domain costs a positive energy (Appendix A). Since $F(3) > F(n \geq 4)$ for every $r \geq 0$, no structure with domains longer than three is ever stable at this order. Comparing FM, $\langle 3 \rangle$ and $\langle 2 \rangle$ gives the phase boundaries

$$\kappa_{\text{FM}|3} = \frac{1}{2} - \frac{g^2}{8} F(3), \quad \kappa_{3|2} = \frac{1}{2} + \frac{g^2}{4} F(3). \quad (9)$$

Quantum fluctuations therefore open a wedge of $\langle 3 \rangle$ of width

$$\Delta \kappa = \frac{3g^2}{8} F(3) = \frac{9g^2}{4(2 + r)(3 + r)(5 + r)} \quad (10)$$

that straddles $\kappa = 1/2$, with twice as much of it on the antiphase side. For decoupled layers, $F(3) = 1/5$ and the wedge is $1/2 - g^2/40 < \kappa < 1/2 + g^2/20$. For $r = 1$, $F(3) = 1/12$ and the wedge is $1/2 - g^2/96 < \kappa < 1/2 + g^2/48$, narrower by the factor $72/30 = 2.4$. The interlayer coupling adds the same constant r to every local field, which compresses the differences between $1/h$ values that drive the selection.

Two features of Eq. (9) bear on what happens at higher order. At the boundary $\kappa_{\text{FM}|3}$ a domain longer than three costs a finite energy, so the transition between the ferromagnet and $\langle 3 \rangle$ is a simple first-order level crossing. At $\kappa_{3|2}$, every sequence of two- and three-site domains has the same energy at order g^2 , so higher orders decide whether mixed structures such as $\langle 2^k 3 \rangle$ appear there, as they do in the thermal problem [3]. [Decide whether to carry the expansion to order g^4 at the $\langle 3 \rangle|\langle 2 \rangle$ boundary, or state it as open.]

For fixed $\kappa > 1/2$, Eq. (9) predicts $\langle 3 \rangle$ above $g_{3|2} = [4(\kappa - 1/2)/F(3)]^{1/2}$. At $\kappa = 0.5625$ this is $g_{3|2} = 1.12$ for $r = 0$ and 1.73 for $r = 1$; at $\kappa = 0.75$ it is 2.24 and 3.46. The last value probably lies beyond the paramagnetic boundary, in which case $\langle 3 \rangle$ never appears at $\kappa = 0.75$ in the bilayer. The perturbative values are only indicative at $g \gtrsim 1$, where $g/[2(2 + r)]$ is no longer small; Sec. V tests them.

C. Fourth and sixth order

At $\kappa_{3|2}$ every sequence of two- and three-site domains has the same energy at order g^2 , so higher orders decide which structures are stable there. We carry the expansion to order g^6 . Moving a wall takes a whole row of flips, so structures with different walls mix only at order g^{L_x} (g^{2L_x} for $r > 0$). For $L_x \rightarrow \infty$ each straight-wall structure is therefore a nondegenerate reference state, and Rayleigh–Schrödinger theory applies to each one separately. We write its energy per spin as

$$e = e_0 + \varepsilon_2 g^2 + \varepsilon_4 g^4 + \varepsilon_6 g^6 + O(g^8), \quad (11)$$

where $\varepsilon_2 g^2$ is the term e_2 of Eq. (6). Odd orders vanish because every flipped spin must be flipped back. Write

TABLE II. Coefficients of Eq. (11) per spin at $\kappa = 1/2$, in units of J_0 , for the structures that compete near $\kappa_{3|2}$; ρ is the wall density. $W_4 = k\varepsilon'_2 + \varepsilon_4$ with $k = F(3)/4$ [Eq. (14)] is the fourth-order energy on the $\langle 3 \rangle|\langle 2 \rangle$ line (not needed for FM). ε_2 and ε_4 are exact; W_4 and ε_6 are rounded.

r		ρ	ε_4	$10^3 W_4$	$10^5 \varepsilon_6$
0	FM	0	-5/1512	—	-24.15
	$\langle 3 \rangle$	1/3	-1/125	-7.333	-7.454
	$\langle 23 \rangle$	2/5	-54673/7560000	-4.610	-120.5
	$\langle 2 \rangle$	1/2	-13/4320	2.546	-22.06
1	FM	0	-1/1152	—	-2.062
	$\langle 3 \rangle$	1/3	-437/362880	-1.011	4.409
	$\langle 23 \rangle$	2/5	-4477/3628800	-0.5972	-2.175
	$\langle 2 \rangle$	1/2	-11/13440	0.4836	-2.055

the Ising part of Eq. (1) as $-\sum J_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z$, so that $J_{ij} = -\kappa$ for the pairs at distance 2 along y . Let $D_i = 2h_i$ be the cost of flipping spin i , and $B_{ij} = J_{ij}s_i s_j$ minus the energy of the bond between spins i and j . Then

$$N\varepsilon_4 = \sum_i \frac{1}{D_i^3} - \sum_{\langle ij \rangle} \frac{4B_{ij}(D_i + D_j)}{(D_i D_j)^2 (D_i + D_j - 4B_{ij})}. \quad (12)$$

The first term is the g^4 term of an isolated spin, whose energy is $-(h^2 + g^2)^{1/2}$. The second sum runs over coupled pairs only. For an uncoupled pair, the fourth-order processes that flip both spins cancel exactly against the normalization term of the expansion. This cancellation generalizes. Each spin that takes part in a process is flipped and restored, so the g^{2n} coefficient receives contributions only from connected clusters of at most n spins. We obtain ε_6 from this linked-cluster form, solving every connected cluster of up to three spins exactly in rational arithmetic. Equation (12) agrees with exact diagonalization of small clusters to 3×10^{-6} relative. The sixth-order coefficient agrees with exact diagonalization to 10^{-4} relative, and with an independent evaluation that does not use the cluster decomposition to 10^{-15} . Table II lists the coefficients at $\kappa = 1/2$.

Expansion about the multiphase point. Write $\kappa = \frac{1}{2} + kg^2 + k_1 g^4 + k_2 g^6$. The classical energy (3) is linear in κ with slope $1 - 4\rho$, so Eq. (11) becomes

$$\begin{aligned} e = e_0|_{\kappa=1/2} &+ g^2[k(1 - 4\rho) + \varepsilon_2] \\ &+ g^4[k_1(1 - 4\rho) + W_4] \\ &+ g^6[k_2(1 - 4\rho) + W_6] + O(g^8), \end{aligned} \quad (13)$$

with the κ derivatives (primes) taken at $\kappa = 1/2$:

$$W_4 = k\varepsilon'_2 + \varepsilon_4, \quad W_6 = k_1\varepsilon'_2 + \frac{1}{2}k^2\varepsilon''_2 + k\varepsilon'_4 + \varepsilon_6. \quad (14)$$

On the second-order boundary, $k = F(3)/4$, the g^2 bracket is the same for every sequence of two- and three-site domains (Appendix A). Each domain of $n \geq 4$ sites adds $\frac{1}{2}g^2[(n - 2)F(3) - F(n)] > 0$ to the numerator of

Eq. (7). At order g^4 only the sequences of 2s and 3s therefore compete, with energy $k_1(1 - 4\rho) + W_4$ per spin. Since the first term is linear in ρ , the stable structures are the vertices of the lower convex hull of W_4 plotted against ρ .

Fourth order: the $\langle 23 \rangle$ phase. At $\kappa = 1/2$ the field (5) of a spin depends only on its position in its own domain (Table I). The pairs in Eq. (12) lie either within one domain or on the two sides of a single wall. W_4 is therefore a sum of terms for single domains and for pairs of adjacent domains [Eq. (A2)]. A periodic sequence can be read as a closed path through the domain lengths, and its energy per spin is the cost of the path divided by its length. Such a ratio is smallest on an elementary cycle, so only $\langle 2 \rangle$, $\langle 3 \rangle$ and $\langle 23 \rangle$ ($\uparrow\uparrow\downarrow\downarrow$) can be stable at this order, among sequences of any length (Appendix A). Table II places $\langle 23 \rangle$ below the chord from $\langle 3 \rangle$ to $\langle 2 \rangle$, by 619/504000 per spin at $r = 0$ and 667/3628800 at $r = 1$. $\langle 23 \rangle$ is therefore stable in a window

$$\frac{1}{2} + \frac{1}{4}F(3)g^2 + k_1^- g^4 < \kappa < \frac{1}{2} + \frac{1}{4}F(3)g^2 + k_1^+ g^4, \quad (15)$$

with $k_1^- = 20591/2016000 \approx 0.0102$ and $k_1^+ = 6011/336000 \approx 0.0179$ at $r = 0$, and $k_1^- = 167/107520 \approx 0.00155$ and $k_1^+ = 1961/725760 \approx 0.00270$ at $r = 1$. The width is

$$\begin{aligned} \Delta\kappa_{23} &= \frac{619}{80640} g^4 \approx 0.00768 g^4 \quad (r = 0), \\ &= \frac{667}{580608} g^4 \approx 0.00115 g^4 \quad (r = 1). \end{aligned} \quad (16)$$

The bilayer window is 6.7 times narrower. This is a much stronger effect of the interlayer coupling than the factor 2.4 for the $\langle 3 \rangle$ wedge. The window keeps shrinking with r : its width is $0.00274 g^4$ at $r = 1/2$ and $0.00028 g^4$ at $r = 2$. The metastable branches $\langle 3 \rangle$ and $\langle 2 \rangle$ cross inside the window, at $\kappa = \frac{1}{2} + \frac{1}{4}F(3)g^2 + k_1^{32}g^4$ with $k_1^{32} = 1067/72000$ ($r = 0$) and $31/13824$ ($r = 1$). This is the line along which template free energies of $\langle 3 \rangle$ and $\langle 2 \rangle$ can be compared with simulations. On the ferromagnetic side the boundary moves to $\kappa_{\text{FM}|3} = \frac{1}{2} - \frac{1}{8}F(3)g^2 + k_1^{\text{F}}g^4$, with $k_1^{\text{F}} = -59/10080$ ($r = 0$) and $-1573/1935360$ ($r = 1$). No new structure appears there. At $\kappa_{\text{FM}|3}$ every structure other than the ferromagnet and $\langle 3 \rangle$ contains two-site domains or domains of four or more sites. Each of these costs energy at order g^2 , which higher orders cannot overturn at small g .

Degenerate edges. The argument also shows exactly which structures tie on the window edges. In the (ρ, W_4) plane, a sequence with no two adjacent three-site domains ($\langle 223 \rangle$, $\langle 2223 \rangle$, $\langle 22323 \rangle$, ...) lies exactly on the segment joining $\langle 2 \rangle$ and $\langle 23 \rangle$. A sequence with no two adjacent two-site domains ($\langle 233 \rangle$, $\langle 2333 \rangle$, ...) lies exactly on the segment joining $\langle 23 \rangle$ and $\langle 3 \rangle$ (Appendix A). Each edge of the window is therefore itself a multiphase line at order g^4 , and the next order decides it.

Sixth order. A connected cluster of three spins spans at most one intermediate domain. W_6 is therefore a sum of terms for three consecutive domains, which we have

confirmed in exact arithmetic. Applied to pairs of consecutive domains, the elementary-cycle argument allows at most one new structure in each gap at this order: $\langle 223 \rangle$ between $\langle 23 \rangle$ and $\langle 2 \rangle$, or $\langle 233 \rangle$ between $\langle 3 \rangle$ and $\langle 23 \rangle$. $\langle 223 \rangle$ opens, in a window of width $4.1 \times 10^{-4} g^6$ at $r = 0$ and $2.3 \times 10^{-5} g^6$ at $r = 1$. $\langle 233 \rangle$ does not open. It lies above the segment from $\langle 3 \rangle$ to $\langle 23 \rangle$ by $2.0 \times 10^{-6} g^6$ per spin at $r = 0$ ($1.2 \times 10^{-7} g^6$ at $r = 1$), a margin small enough that the next order could reverse it. Through order g^6 the transverse field thus orders the structures as $\langle 3 \rangle$, $\langle 23 \rangle$, $\langle 223 \rangle$, $\langle 2 \rangle$ with increasing κ . We do not follow the sequence further. **[Compare with the order in which $\langle 2^k 3 \rangle$ phases appear in the low-temperature expansion of the classical model [3] and in Fisher–Szpilka; state it only after checking the papers.]**

The sixth order also shifts the edges of the $\langle 23 \rangle$ window. With $k_2^- = -3.06 \times 10^{-3}$ and $k_2^+ = +5.26 \times 10^{-3}$ at $r = 0$, and -1.43×10^{-4} and $+2.42 \times 10^{-4}$ at $r = 1$ (the upper edge is now the $\langle 23 \rangle | \langle 223 \rangle$ boundary),

$$\begin{aligned} \Delta\kappa_{23} &= 0.00768 g^4 (1 + 1.08 g^2) & (r = 0), \\ &= 0.00115 g^4 (1 + 0.34 g^2) & (r = 1). \end{aligned} \quad (17)$$

The correction is 27% of the leading term at $g = 0.5$ for $r = 0$, and it equals the leading term near $g = 1$. Equation (16) is therefore quantitative only at small g . Where the window becomes wide enough to resolve numerically, the sixth-order terms are needed, and at $r = 0$ the series converges slowly there.

D. Why the square lattice differs from the chain

The perturbative argument relies on the walls being rigid. On the square lattice, moving a wall by one site requires flipping a whole row of L_x spins, which enters only at order g^{L_x} , so the walls are frozen at any finite order. In the chain ($J_x = 0$, $r = 0$) the field next to a wall vanishes at $\kappa = 1/2$ by Eq. (5), a single flip moves the wall at no cost, and the walls delocalize, as described by effective domain-wall Hamiltonians for the transverse chain [11]. The large-spin analysis of an anisotropic Heisenberg model [16, 17] also finds that quantum fluctuations lift the degeneracy, but by a different mechanism, spin deviations of large spins rather than transverse-field flips, and with a different result. At the simulation temperature $T = 0.15 J_0$, thermal steps on a wall cost at least $2J_0$ each and are suppressed by a factor of order $e^{-2J_0/T} \approx 10^{-6}$, so the walls are straight on the scale of our lattices. **[Check this estimate against the measured wall roughness.]**

IV. METHODS

A. Stochastic series expansion

We write Eq. (1) as $\hat{H} = -\sum_b \hat{H}_b - \sum_s (\hat{H}_{s,c} + \hat{H}_{s,f}) + C$, with bond operators $\hat{H}_b = |J_b| + J_b \hat{\sigma}_i^z \hat{\sigma}_j^z$, where J_b is the coupling of bond b in the convention $-J_b \hat{\sigma}_i^z \hat{\sigma}_j^z$, a constant site operator $\hat{H}_{s,c} = \Gamma$, a spin-flip operator $\hat{H}_{s,f} = \Gamma \hat{\sigma}_s^x$, and $C = \sum_b |J_b| + NT$. All matrix elements are nonnegative for either sign of J_b , so frustration affects only the efficiency. Diagonal updates insert and remove $\hat{H}_b$ and $\hat{H}_{s,c}$ in the operator string, and cluster updates build Swendsen–Wang-type clusters that pass through bond operators and end on site operators [27]. The energy is $e = (-\langle n \rangle / \beta + C) / N$ with n the number of operators, the transverse magnetization is $m_x = \langle n_f \rangle / (\beta \Gamma N)$ with n_f the number of flip operators, and the nearest-neighbor, next-nearest-neighbor and interlayer correlations follow from the numbers of bond operators of each type. Equal-time quantities are measured on the propagated state at imaginary time zero. There is no Trotter error.

B. Equilibration in the modulated region

At small g the operator string holds few site operators, the clusters of satisfied bonds percolate, and the cluster update reduces to global flips. Local moves then cannot convert one ordered structure into another. Annealing downward in g from the paramagnet removes most of these traps: on 12-site test clusters, direct runs at $\kappa = 0.75$, $r_\perp = -1$, $g = 0.5$ reach the exact energy in 14 of 24 chains, and annealed runs in 24 of 24. On larger lattices annealing fails in the modulated region. On a 12×12 bilayer at $\kappa = 0.75$, all chains annealed from $g = 3$ to $g = 0.05$ end in $\langle 3 \rangle$ at $e = -1.5837$ for $r = 0$, while the exact low-field energy of the stable antiphase is $e_0 = -1.75$; chains started in $\langle 2 \rangle$ stay there, at $e = -1.7504(2)$. The anneal selects the modulation that forms at the paramagnetic transition, and the lock-in to $\langle 2 \rangle$ at lower field would require restructuring every wall at once. **[Update with the $L = 24$ low-field check.]**

We therefore treat each candidate ordered structure as a separate branch. Chains are started in the FM, $\langle 2 \rangle$ and $\langle 3 \rangle$ structures at $g_0 = 0.05$ and stepped upward in g , and a separate set is annealed downward from $g = 5$. Along each branch the free energy per spin follows from the exact relation $\partial F / \partial \Gamma = -\langle \sum_s \hat{\sigma}_s^x \rangle$,

$$f_b(g) = f_b(g_0) - \int_{g_0}^g m_x^{(b)}(g') dg', \quad (18)$$

where $f_b(g_0) = e_b(g_0)$ because the entropy of an ordered state at g_0 and $T = 0.15 J_0$ is negligible: the lowest excitation costs $2(2+r)J_0$, a Boltzmann factor below 10^{-11} . The integral is valid while the branch remains in one

structure. We cut a chain at the first change of its dominant wavevector. The stable state at each g is the branch of lowest f_b , a crossing of two branches locates a first-order transition, and the range over which both branches persist measures the hysteresis. Before the first cluster update, the operator string is filled by diagonal updates alone so that the initial structure survives (Appendix B).

C. Validation

We benchmark the SSE code against exact diagonalization at finite temperature on four 12-spin periodic clusters built by the production lattice code, so that both methods use the same bond list. The clusters are a 2×3 bilayer, 2×6 and 3×4 single layers, and a 1×6 two-leg ladder, and the parameters include frustrated points up to $\kappa = 1.3$, both signs of $r_\perp$, $J_y < 0$, and $T = 0.15$ and $0.5 J_0$ (Table III). Over 471 comparisons of the energy, m_x , both correlation estimators, template moments and the structure factor, the median $|z|$ is 0.54 and 98.5% of the $|z|$ values are below 2, against 0.69 and 94.3% expected for the Student- t distribution with 23 degrees of freedom of our 24-chain averages; the largest $|z|$ is 2.62. Two checks on large lattices are the low-field energies, which must approach $e_0 = \min(-2 + \kappa, -1 - \kappa) - r/2$ from below by $O(g^2)$, and the two-dimensional transverse-field Ising model ($\kappa = r_\perp = 0$), whose finite-temperature transition is known accurately [31, 32]. [Low-field check at $L = 12$ and 24; transverse-field Ising crossings at $T = 0.15$ and with $\beta = L$.]

V. RESULTS

A. Branch free energies near the multiphase point

[$f_b(g)$ for the FM, $\langle 2 \rangle$, $\langle 3 \rangle$ and annealed branches at $\kappa = 0.5625$ and 0.75 , $r = 0$ and 1 , $L = 12$; crossings g^* with bootstrap intervals; hysteresis windows; comparison with $g_{3|2}$ from Eq. (9).]

B. The $\langle 3 \rangle$ wedge and the interlayer coupling

[Wedge boundaries from κ cuts at three or more fields, for $r = 0$ and 1 ; finite-size check including lattices with L_y not divisible by 4 or 6; comparison of the measured width with Eq. (10) and of the measured narrowing with the factor 2.4.]

C. Paramagnetic boundary

[$g_c(\kappa)$ at $r = 0$ and 1 from U_4 and ξ/L crossings on at least four sizes, with drift of the crossings; the field range extended to $g \approx 5$ for $r = 1$.]

D. Modulated region beyond the wedge

[$g^*(g)$ at $\kappa = 0.5$, 0.5625 and 0.625 , a scan in L_y , and bounds on any incommensurate or floating region between the commensurate phases and the paramagnet. Resolving whether a floating phase separates $\langle 3 \rangle$ from the paramagnet requires far larger lattices and is left open.]

VI. DISCUSSION AND CONCLUSION

For spin $1/2$ in a transverse field on the square lattice, quantum fluctuations select $\langle 3 \rangle$ at the multiphase point at order g^2 . The mechanism is local: spins next to a straight domain wall have the smallest molecular field and gain the most energy from virtual flips, and domains of three sites carry the largest number of such spins per site among domains that avoid the two-site penalty. The interlayer coupling raises every local field by the same amount and weakens the selection. [Quantitative comparison with the simulations; the paramagnetic boundary; remaining open questions, including the $\langle 3 \rangle | \langle 2 \rangle$ boundary at higher order and a floating phase at larger field.]

The equilibration problem is not specific to our algorithm. Any protocol that crosses the paramagnetic boundary of the modulated region, whether thermal or quantum annealing, inherits the modulation of the ordering instability and needs a collective rearrangement to reach a different commensurate state at lower field. [Decide whether to keep this remark on annealers.]

ACKNOWLEDGMENTS

[Acknowledgments.]

DATA AVAILABILITY

[Repository of the SSE code, the analysis scripts and the data; confirm that the repository is public.]

Appendix A: Minimum of the domain-sequence energy

Write the per-domain cost as $c(n) = (2 - 4\kappa) - \frac{g^2}{2} F(n)$, so that $\Delta e = \sum_k c(n_k) / \sum_k n_k$ by Eq. (7). For positive n_k , $\sum_k c(n_k) / \sum_k n_k \geq \min_k c(n_k) / n_k$, with equality when every domain has the same ratio $c(n_k) / n_k$. If $c(n) > 0$ for all n , the minimum is the ferromagnet, $\Delta e = 0$. Otherwise it is reached by the periodic structure that minimizes $c(n)/n$. From Eq. (8),

$$F(3) - F(n \geq 4) = \frac{2}{(3+r)(4+r)(5+r)} > 0. \quad (\text{A1})$$

With F constant for $n \geq 4$, $\langle n \rangle$ with $n \geq 4$ can beat both the ferromagnet and $\langle 3 \rangle$ only if $F(n) > F(3)$, which never holds. The remaining comparisons, FM against $\langle 3 \rangle$ ($c(3) < 0$) and $\langle 3 \rangle$ against $\langle 2 \rangle$ ($c(3)/3 < c(2)/2$), give Eq. (9). At $\kappa_{3/2}$ one has $c(2)/2 = c(3)/3$, and any sequence of two- and three-site domains has the same Δe .

Higher orders. At order g^4 on the line $k = F(3)/4$ the same ratio structure appears, with the domain costs replaced by costs of adjacent domain pairs. Let $P(D, D', B) = -4B(D + D')/[(DD')^2(D + D' - 4B)]$ be the pair term of Eq. (12). Let $D_1, \dots, D_n$ be the flip costs along a domain (Table I), and let $\gamma_i = s_i(s_{i+2} + s_{i-2})$, which takes the values $-2, 0$ and 2 . Let X_n and Y_n be the flip costs of the first and second site from a wall of a domain of length n . Then, for a period of L sites,

$$\begin{aligned}
W_4 L &= \sum_k [a(n_k) + b(n_k, n_{k+1})], \\
a(n) &= \sum_{i=1}^n \left[-\frac{2k\gamma_i}{D_i^2} + \frac{1}{D_i^3} \right] \\
&\quad + \sum_{i=1}^n \left[P(D_i, D_i, 1) + \frac{1}{2}P(D_i, D_i, r) \right] \\
&\quad + \sum_{i=1}^{n-1} P(D_i, D_{i+1}, 1) + \sum_{i=1}^{n-2} P(D_i, D_{i+2}, -\frac{1}{2}), \\
b(n, n') &= P(X_n, X_{n'}, -1) + P(Y_n, X_{n'}, \frac{1}{2}) \\
&\quad + P(X_n, Y_{n'}, \frac{1}{2}). \tag{A2}
\end{aligned}$$

With $C(n, n') = k_1(n - 4) + a(n) + b(n, n')$, the fourth-order energy per spin is $\sum_k C(n_k, n_{k+1}) / \sum_k n_k$. A periodic sequence is a closed walk on the directed graph with vertices $\{2, 3\}$ and all four edges, where edge $n \rightarrow n'$ has cost $C(n, n')$ and length n . Every closed walk splits into elementary cycles. A ratio of sums with positive denominators is at least the smallest ratio of its parts, with equality only if all parts share that ratio. The elementary cycles are $2 \rightarrow 2$, $3 \rightarrow 3$ and $2 \rightarrow 3 \rightarrow 2$, that is $\langle 2 \rangle$, $\langle 3 \rangle$ and $\langle 23 \rangle$, so one of them minimizes the energy for every k_1 .

Now count the adjacent pairs N_{22} , N_{33} and $N_{23} = N_{32}$ of a cyclic sequence. The period is $L = 2N_{22} + 3N_{33} + 5N_{23}$, and the total cost is $N_{22}C(2, 2) + N_{33}C(3, 3) + N_{23}[C(2, 3) + C(3, 2)]$. The point (ρ, W_4) of the sequence is therefore the average of the points of $\langle 2 \rangle$, $\langle 3 \rangle$ and $\langle 23 \rangle$, weighted by the site fractions $2N_{22}/L$, $3N_{33}/L$ and $5N_{23}/L$. Sequences with $N_{33} = 0$ lie on the segment from $\langle 2 \rangle$ to $\langle 23 \rangle$, and sequences with $N_{22} = 0$ lie on the segment from $\langle 23 \rangle$ to $\langle 3 \rangle$. All other sequences lie strictly above the hull, because $\langle 23 \rangle$ lies strictly below the chord from $\langle 3 \rangle$ to $\langle 2 \rangle$.

At order g^6 the terms of Eq. (14) involve at most three consecutive domains. The same argument, applied to the graph whose vertices are pairs of consecutive domains, leaves $\langle 2 \rangle$, $\langle 23 \rangle$ and $\langle 223 \rangle$ as the only candidates on the $\langle 23 \rangle|\langle 2 \rangle$ line, and $\langle 3 \rangle$, $\langle 23 \rangle$ and $\langle 233 \rangle$ on the $\langle 3 \rangle|\langle 23 \rangle$ line.

Appendix B: SSE implementation and exact-diagonalization benchmark

Bonds are grouped into classes with a common $|J_b|$ (bonds along x , bonds along y , next-nearest-neighbor bonds along y , rungs). An insertion first chooses a class with probability proportional to its total weight and then a bond uniformly within it, and is rejected when the chosen bond is unsatisfied. The string length grows as $M = n + n/2 + 64$ during thermalization. Each chain starts with 200 diagonal-only sweeps at its first field. Without them, the first cluster update on a nearly empty string would flip every operator-free spin with probability $1/2$ and randomize the initial structure. Diagonal updates cannot change the spins, so this step is part of the thermalization and does not affect the sampled distribution.

The benchmark of Table III anneals every case over 16 fields from $g = 3$ to the target, with 2000 sweeps per field, followed by 5000 thermalization and 40 000 measurement sweeps at the target, in 24 independent chains.

TABLE III. SSE against exact diagonalization on 12-spin periodic clusters. e is the energy per spin and m_x the transverse magnetization, both in units of J_0 ; the SSE values are means of 24 chains with standard errors in parentheses. The last column is the largest $|z|$ over all observables of the case (31–39 per case).

Cluster	T/J_0	J_y/J_0	κ	$r_\perp$	g	e (ED)	e (SSE)	m_x (ED)	m_x (SSE)	$\max z $
2×3 bilayer	0.15	1	0	1	1.0	-2.60042	-2.60075(20)	0.20169	0.20176(14)	1.65
2×3 bilayer	0.15	1	0.75	-1	0.5	-1.78588	-1.78603(17)	0.14416	0.14394(23)	2.01
2×3 bilayer	0.50	1	0.5	1	2.0	-2.52091	-2.52076(68)	0.54569	0.54629(57)	2.62
2×6 layer	0.15	1	0	0	2.0	-2.52595	-2.52599(23)	0.56144	0.56135(18)	0.68
2×6 layer	0.15	1	0.75	0	1.0	-1.83990	-1.84065(41)	0.46098	0.46162(69)	1.83
2×6 layer	0.15	1	0.5625	0	1.5	-2.07157	-2.07171(41)	0.64328	0.64283(47)	1.86
2×6 layer	0.15	-1	0.25	0	1.0	-1.89600	-1.89616(20)	0.29902	0.29881(20)	1.82
2×6 layer	0.50	1	1.0	0	0.5	-1.71530	-1.71590(69)	0.18847	0.18876(157)	2.00
1×6 ladder	0.15	1	0.75	1	1.0	-1.46156	-1.46128(36)	0.61156	0.61140(57)	1.80
1×6 ladder	0.15	1	1.3	-1	0.3	-1.30261	-1.30361(86)	0.22459	0.22781(527)	2.50
1×6 ladder	0.15	1	0.5	1	3.0	-3.13579	-3.13617(44)	0.95525	0.95528(12)	1.68
3×4 layer	0.15	1	0.375	0	2.5	-2.74079	-2.74034(26)	0.88540	0.88520(14)	1.68
3×4 layer	0.15	1	1.0	0	1.0	-2.12664	-2.12642(20)	0.25671	0.25629(20)	2.08

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